

Tag & Track Sprint 1

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1. Introduction

- Introductions - Tag and Track Team
 - Martin Hundrup - Team Lead and CS Major, Back End
 - Sarah Carley - CS Major, Front End
 - Shawn Will - CS Major, Database
- Objectives
 - Gather requirements.
 - Create appropriate documentation.
 - Identify and learn potential frameworks.
 - Come up with a solid idea of what we would like our project to look like.

2. Sprint Objectives

- Collect requirements from the client.
- Create use cases, user stories, and a UML diagram.
- Evaluate the existing system and think on how to tailor our project to improve this.
- Identify appropriate development tools and start learning backend/frontend information that may be needed.
- Start development if possible given the time span and other factors.

3. Requirements Gathering Process

Meetings:

- We have met with our client multiple times to gain more understanding on what kind of application they would like
- We have gone on an in-person tour to better see the current solution and know the environment we will be developing for.
- We have had a few meeting so far as seen: [MoM1](#) [MoM2](#)

3.1 Requirements Gathering Process

Key Requirements:

- Our client wants an application that supports a local database of specimens that can keep track of whether or not a specimen is checked out.
- We want to be able to generate and scan QR codes to change a specimen's check in/out status and create loans.
- We want the QR code to also be able to show specimen information pulled from the museum's existing database.

4.1 User Stories

US-03:

As a user, I want to easily scan QR codes so I can view specimens.

Acceptance Scenarios (Gherkin):

Scenario 1: Positive Flow

Given a QR code that references a database entry

When the QR code is scanned

Then the database entry should be displayed to the screen

Scenario 2: Negative/Edge Case

Given a QR code that does not reference a database entry

When the QR code is scanned

Then an error should be thrown and the user should be notified of the bad QR code read

4.2 User Stories

US-06:

As a user, I want to be able to see the check history of an item so I can track the item.

Acceptance Scenarios (Gherkin):

Scenario 1: Positive Flow

Given a specimen database entry with a check history

When the user scans its QR code or searches for the item

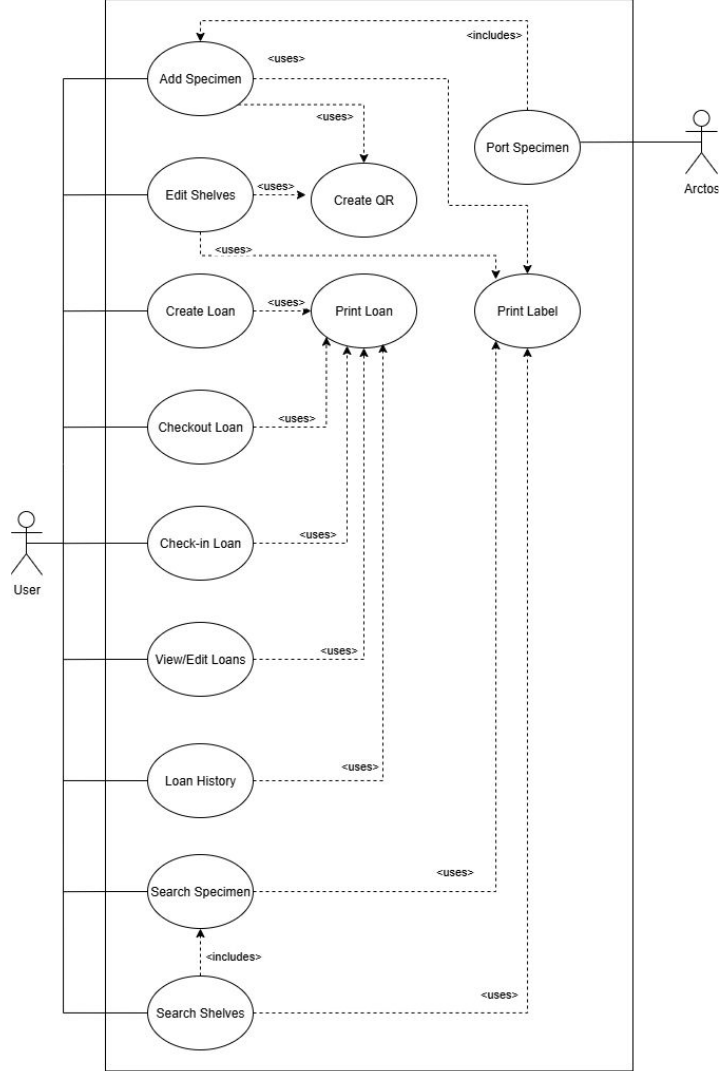
Then the running check in/out history should be displayed to the screen

Scenario 2: Negative/Edge Case

Given a specimen database entry without a check history

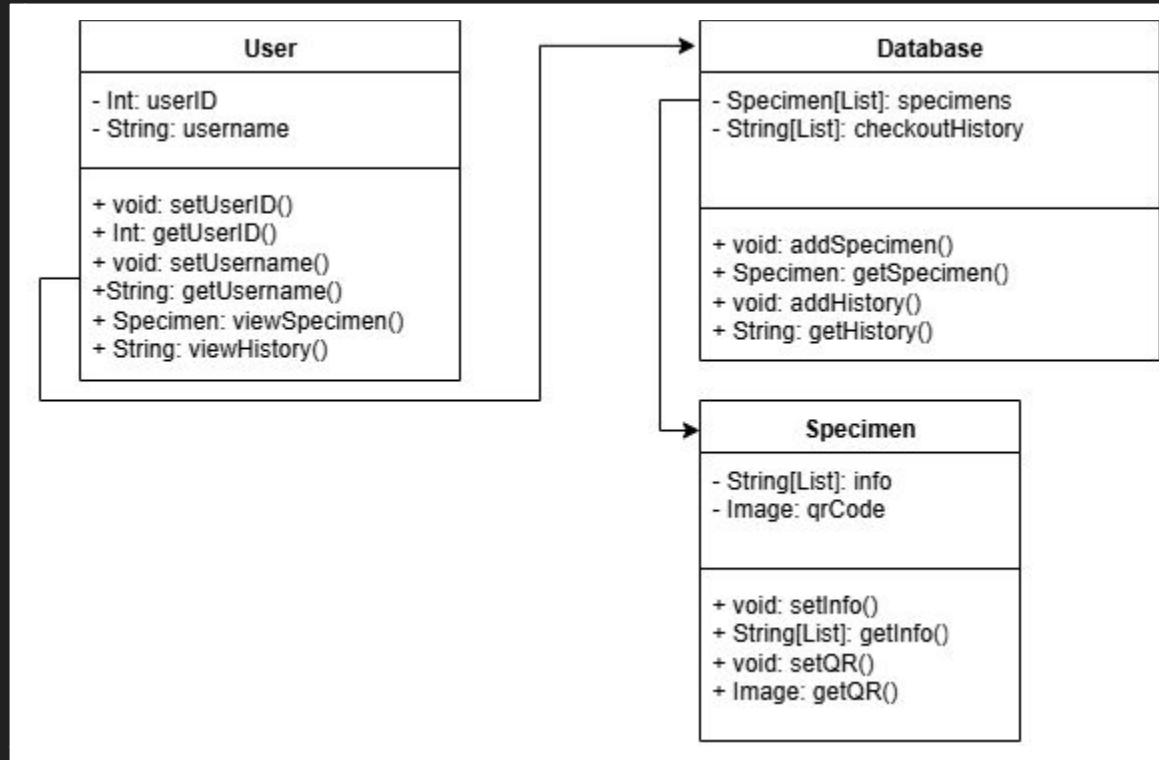
When the user scans its QR code or searches for the item

Then the screen should alert the user that the item has no check history

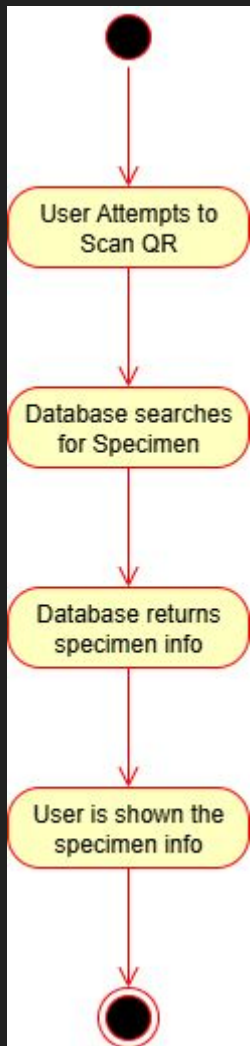


4.3 Use Case Diagram

4.4 UML Class Diagram



4.5 UML Activity Diagrams



5. Evaluation of Existing Solution

New loans currently have a lengthy process:

1. The employee gets a loan request from someone.
2. Based on this request, the employee goes to each specimen and writes all of them down by hand.

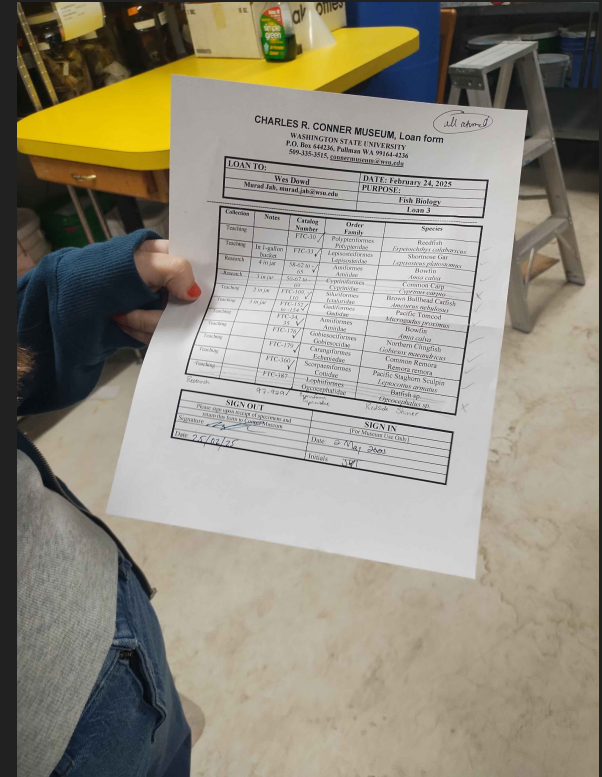
Shown on right: what some of the specimen storage looks like



5.1 Evaluation of Existing Solution

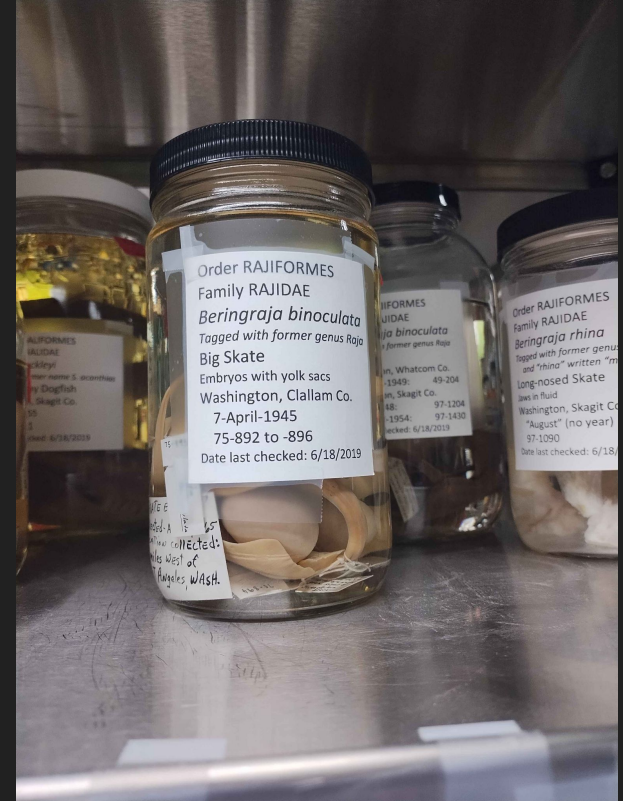
3. After writing down all of the specimens for the loan, the list is then typed out onto the computer.
4. This list is sent to the requester for approval.
5. Once approved, the employee returns back to the specimens with a loan sheet associated with the current loan.

Shown on right: an example of a loan sheet.



5.2 Evaluation of Existing Solution

6. The employee uses the loan sheet as a “shopping list” and finds all of the specimens based on the list.
7. The requester then will come to pick up the loan.
8. Both the employee and requester will sign the loan sheet to confirm pickup/confirmation of when the loan is due back.
9. Employee then manually updates a spreadsheet with the statuses of all of the loan specimens being checked out.



5.3 Evaluation of Existing Solution

Overall Evaluation

- While this current solution works, it is extremely time consuming. Depending on the number of specimens in a loan, this process could take up to two hours (not counting the time it takes for requester's approval).
- The process of creating a loan by hand, then transferring to a digitized document in itself can be lengthy and is inefficient considering how the museum has over 60,000 specimens.
- We aim to reduce the time it takes to complete this process by at least 50% by giving a quick and easy way to create a new loan and track check in/out information.

6. Kanban Overview

Kanban Board

7. Feature Demonstration

Prototype

8. Skills Identification and Learning Process (Martin)

- **Skill Set Identified:**
 - I needed to familiarize myself with iPad application development, and coding interactive UI applications in C#
- **Learning Resources:**
 - I have experience coding interactive UI in C# through past classes (321, 327), as well as experience learning a new UI framework in Go (for an internship)
 - I did research into .NET MAUI for Mac/iPad OS development from Windows, which utilizes C#.
- **What I Learned:**
 - Our development workflow, the style of UI we'll be making, and that I hate developing for Apple.

8.1 Skills Identification and Learning Process (Sarah)

- **Skill Set identified:**
 - I needed to learn more about applications with user interfaces.
- **Learning Resource:**
 - I went through many different parts of the .NET documentation.
- **What I learned:**
 - I learned how to connect a frontend and backend easier and how to support low coupling a high cohesion with event-driven applications. I also gained a lot of C# experience.

8.2 Skills Identification and Learning Process (Shawn)

- **Skill Set identified:**
 - I need to better understand how mobile development works.
- **Learning Resource:**
 - I researched into what development methods we could use to create an iPad app, mainly .NET MAUI.
- **What I learned:**
 - I learned that Apple is the worst company to develop anything for due to their restrictions and all the hoops you have to jump through.

9. Sprint Achievements and Challenges

Achievements

- Requirements Gathered
- Documentation created
- Process evaluation
- Skills learned
- Interactive prototype created

Challenges

- Starting development due to device issues
- Apple

10. Responsible Use of AI

Disclosure

- Some AI (Clippy) was used during this Sprint.

Application and improvements

- Clippy aided in development workflow research (way to develop for iPad/Mac OS from Windows) and to help edit documentation for increased readability.

Limitations and Checks

- We had Clippy cite all of its sources, and we reviewed them manually.

We understand that final decisions and deliverables are our responsibility.

11. Next Step and Retrospective

Next steps:

- Start coding the frontend and backend.
- Create a rough database until we can interface with Arctos API.
- Plan out when we'll need access to a Mac
- Get an Apple development account

Retrospective:

What went well

- Requirements gathering
- Meetings with client
- Document creation
- User Interface prototyping

What went poorly

- Deciding a framework
- Finding a suitable Apple device for development and permissions

Thank you